

Creating Web Clipping Applications

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Conference 2000

Web Clipping Overview

- Web Clipping Introduction
- The System
- Web Clipping Development

What Is a Web Clipping Application?

- Set of web pages hosted on Palm OS® devices
  - Palm database
  - Pages and images are records
- Created with simple desktop tools
- Rendered by WCA Viewer (Clipper)
- Capable of:
  - Viewing local pages
  - Sending form queries to remote servers
  - Launching Palm OS® applications

What Is a Web Clipping?

- HTML sent by remote server in response to query from WCA
  - Received on device in WCA format
- Created by remote server
- Rendered by WCA Viewer (Clipper)

Top 5 Reasons to Build Web Clipping Applications

1. WCAs are easy to build
2. The tools for building WCAs are free
3. WCAs leverage existing web infrastructure
4. WCAs provide instant gratification
5. WCAs provide rapid solutions for clients and customers
6. …wireless is wicked cool!

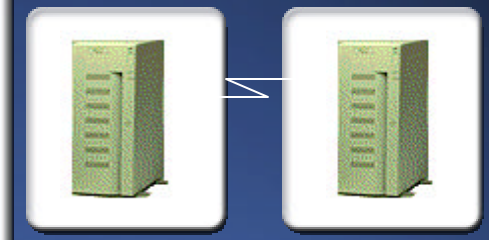
System Components

Hardware



Software

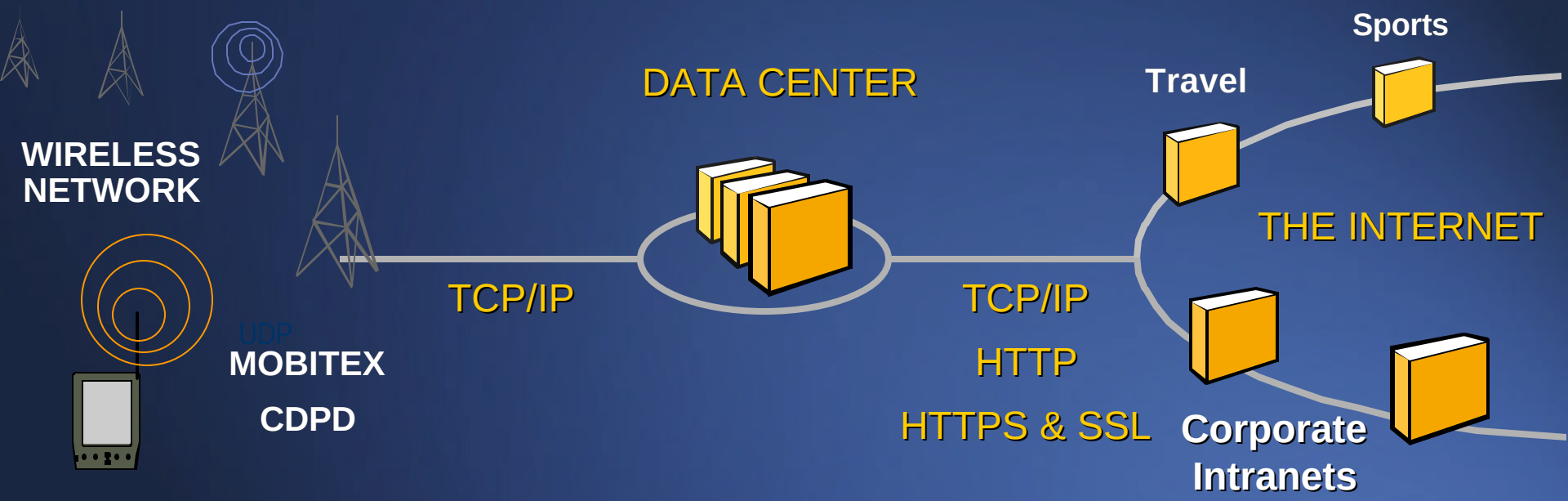




Services

Palm.Net OmniSky

Palm.Net Architecture



Tools of the Trade

- Text editor
- Graphics program
- WCA Builder
- Palm OS® Emulator (POSE)
- Palm OS ROM file
- Web clipping device
  - Palm VII series handhelds
  - Palm V series handhelds with OmniSky modem
  - Palm MIK with compatible device and cellphone or modem

Skill Sets

- HTML
- Server scripting
  - CGI or any other process that returns HTML
  - Best if Palm-Friendly HTML
- Others
  - Web server configuration
  - Firewall configuration

WCA Development Process

1. Design the application
2. Code app in HTML
   Code back-end server
3. Convert HTML to a WCA (.pqa)
4. Install the WCA on a device/POSE
5. Test
6. Debug



Demo: Hello, World!

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Design Guidelines

- Determine target audience
- Determine content
  - Client (WCA)
  - Server (CGI, etc.)
- Think small
  - Screen size 153 x 144
  - Bytes per request and response
  - Scrolling—avoid if possible
    - Main page
    - Clippings

HTML Set Overview

- HTML 3.2, except the following:
  - Applets
  - JavaScript
  - Nested tables
  - Image maps
  - VSPACE attribute
  - <SUB>
  - <SUP>
  - <LINK>
  - <ISINDEX>

HTML Set Overview

- Meta tags
  - PalmComputingPlatform
  - PalmLauncherRevision
  - LocalIcon
  - HistoryListText
- Form input element types
  - DatePicker
  - TimePicker

HTML Set Overview

- Special strings
  - %deviceid
  - %zipcode
  - %location
  - %WCDevCaps
- SmallScreenIgnore tag
- Referencing
  - Relative within the WCA
    - Go
  - Absolute within clippings
    - Go

HTML Set Overview

- Graphics
  - gif, jpg, bmp
  - 1-, 2-, or 4-bit grayscale
  - 8- or 16-bit color
  - Sequential gif animation not supported
- Launcher icons
  - 32x22 pixels for the large
  - 15x9 pixels for the small
- Integration with other apps
  - Palm and palmcall

Meta Tags: *PalmComputingPlatform*

- Palm-friendly indicator
- Needed by WCA files
- Needed by clippings

```
<meta name="PalmComputingPlatform"
      content="true">
```

Meta Tags: *PalmLauncherRevision*

- Version number for the PQA
  - Seen in the Applications Launcher screen, version info view

```
<meta name="PalmLauncherRevision"
      content="1.0">
```

Meta Tags: *LocalIcon*

- Used to include images or html pages not referenced locally
- Placed in WCA, not clippings

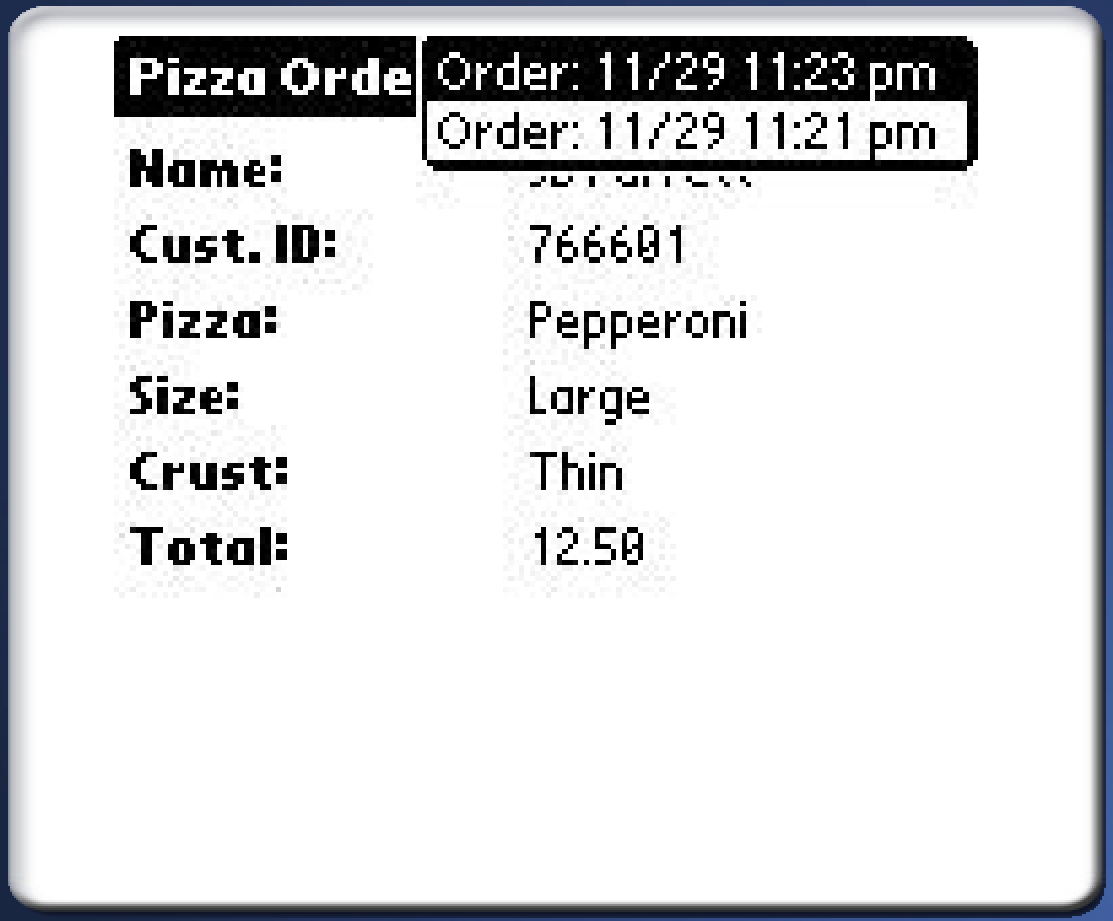
```
<meta name="LocalIcon" content="image.gif">
```

Meta Tags: *HistoryListText*

- Provides informative titles for cached pages
- &date and &time strings in the content attribute

```
<meta name="HistoryListText"
      content="string to display &date &time">
```

History List Example



Form Input Element Types

- DatePicker
  - Displays as set, like MM-DD-YY
  - Displays the current date if value attribute is undefined
  - Sends YYYY-MM-DD format

```
<input type="datepicker"
       name="Date:"
       value="">
```

Form Input Element Types

- TimePicker
  - Displays as set, like HH:MM am/pm
  - Displays the current time if the value attribute is undefined
  - Sends HH:MM 24-hour clock format

```
<input type="timepicker"
       name="Time:"
       value="">
```

Date and Time Pickers Example



Special Strings: %DeviceId

- Two part format:
  - Validation code
    - -1 = invalid device id
    - 0 = we don't know
    - 1 = valid device id
  - Device identifier
    - Maximum of 64 bytes
    - Currently ROM flash ID for non-Palm VII devices
- Can be used in clipping links and form fields

Special Strings: %ZipCode

- Often used for auto-retrieve features
  - Local news, weather, movies, etc.
- Supported for use in clipping links

Special String Examples

```
<form method="get"
      action="http://www.x.com/cgi/x.cgi">
<input type="hidden"
       name="did"
       value="%deviceid">
<input type="hidden"
       name="zip"
       value="%zipcode">
…</form>
<a href="http://www.x.com/x.cgi?did=%deviceid">Go</a>
<a href="http://www.x.com/x.cgi?zip=%zipcode">Go</a>
```

HTML Element: SmallScreenIgnore

- An element used to contain HTML that you don’t want sent over-the-air to a device
  - Content is removed by the clipping proxy server

```
<smallscreenignore>
<img src=“reallyBig.gif” height=“200” width=“300”>
</smallscreenignore>
```

Page References

- Referencing WCA pages from WCA pages:

```
<a href="about.html">About</a>
```

or

```
<a href="about.html" button>About</a>
```

or

```
<img src="cool.gif">
```

Page References

- Referencing WCA pages from clippings:

```
<a href="file:mypqa.pqa/about.html">About</a>
```

or

```
<img src="file:mypqa.pqa/cool.gif">
```

Page References

- Referencing the Internet:
  - `<a href="http://www.palm.com">`
    - Over-the-air symbol automatically included
  - `<a href="https://www.bofa.com">`
    - Secure over-the-air symbol automatically included
  - Over-the-air symbols not included after linked graphics
    - Create as part of graphic

Common Pitfalls—WCAs

- Forgot to include the PalmComputingPlatform meta tag
- Referenced using the file:mine.pqa/about.html syntax from the root page
- Attempting to include an image with a width > than 153 pixels
- Using protocols in a URL other than
  - http, https, file, mailto, palm, palmcall
- Unsupported port # in URL

Common Pitfalls—Web Clippings

- Forgot to include the PalmComputingPlatform meta tag
- Attempted to re-direct to a page stored on the device
  - Note: max of 3 Internet redirects allowed
- Used relative referencing to pages stored on the device
- Sent tables wider than 153 pixels (pre-4.0)
- Sent images wider than 153 pixels
- Using embedded scripting

Common Pitfalls—Back-end Servers

- Forgot to account for the Palm OS® Emulator deviceid of 0.0.0
- Requiring a certificate from the clipping proxy server when establishing an SSL connection
- Server not open to the Internet on the ports web clipping requests are on
  - Note: Use of the emulator requires port 5002 to be open in both directions through the firewall

Development Resources

- Web Clipping Development Page
  - www.palmos.com/dev/tech/webclipping
  - Palm VII White Paper
  - Web Clipping Developer's Guide
  - Web clipping FAQ articles
  - Tutorials
  - Example HTML code
  - WCA Builder
  - Palm OS Emulator

Development Resources

- Development Zone Knowledge Base
  - Oasis.palm.com/dev/kb/
  - Searchable
- Developer Assistance
  - www.palmos.com/dev/tech/support/
  - E-mail forum subscription
  - Newsgroup listings
  - Training info
  - Submit a technical question to Palm DevSupp

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